

GVIM Chemical Intelligence Agent

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Chemical Intelligence Agent

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Project Background

ChatGPT has ushered in the era of large AI models.

-  **OpenAI** company has developed a large-scale language model:
 - It learns language patterns from a massive corpus, enabling it to generate human-like text.
- It functions as a conversational AI capable of understanding natural language and engaging in high-quality multi-turn dialogues.
- –Its human-like responses are astonishing, making it a [milestone breakthrough in artificial intelligence](#).



Bill Gates: The Importance of **ChatGPT** is Comparable to the Invention of the Internet

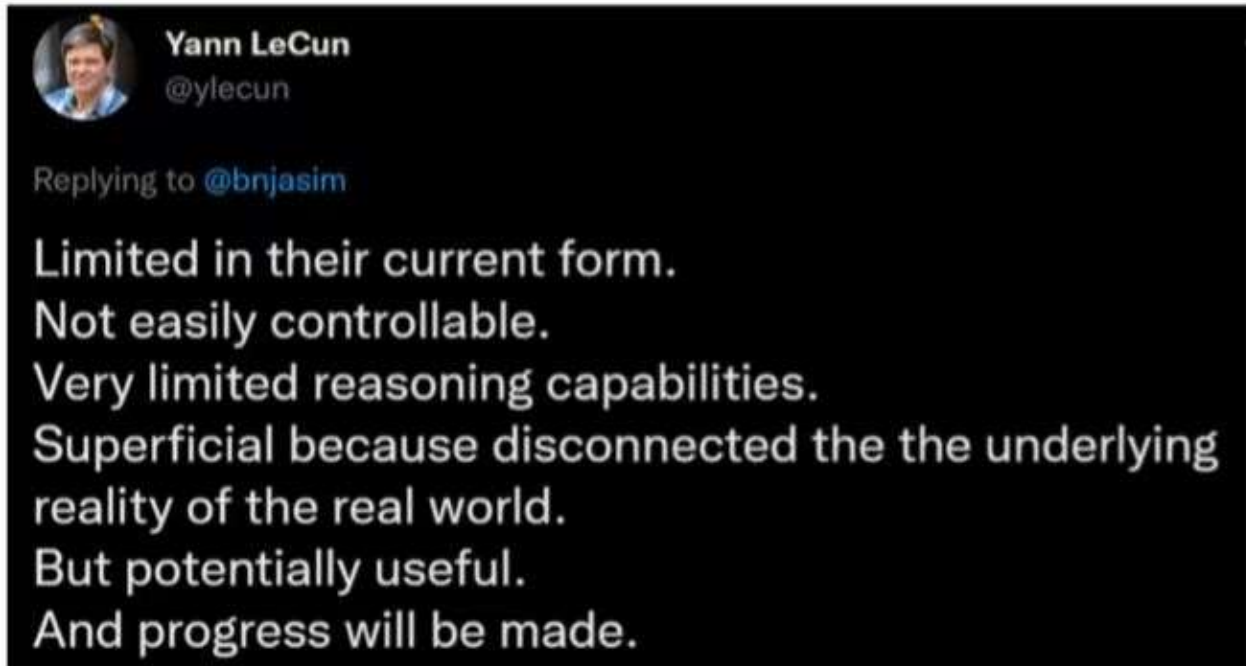
Released on November 30, 2022, it gained 1 million users in just 5 days. By January of this year, its monthly active users had reached 100 million.



Limitations of ChatGPT

- Unable to incorporate new knowledge in real time
- Lacks expertise in highly specialized fields
- Reasoning and computational abilities are still insufficient, leading to possible nonsensical responses
- Only supports text generation
- High computational cost
-

However, most of these issues are engineering challenges and do not present insurmountable theoretical obstacles.



Project Background

Tools Use

To address the **limitations** of large models mentioned above, we can reference how humans effectively handle complex and subtle pattern recognition tasks. Humans can achieve this by supplementing the prior knowledge in their minds with tools like **books, search engines, or calculators**, and only then formulating a conclusion (for example, "The image describes XX").



Project Background

The model not only needs access to external tools but must also be capable of independently planning and executing tasks. A generative AI model with reasoning, logic, and access to external information embodies the concept of an Agent. In other words, an Agent is a program that extends the out-of-the-box capabilities of a generative AI model.

Find restaurants

Model: Plugins ALPHA · Enabled plugins:   



Looking to eat vegan food in San Francisco this weekend. Could you get me one great restaurant suggestion for Saturday and a simple recipe for Sunday (just the ingredients) ? Please calculate the calories for the recipe using WolframAlpha. Finally order the ingredients on Instacart.



Calculate calories



Greens Restaurant - San Francisco, CA on OpenTable

 opentable.com

Used Wolfram|Alpha 

The total calories for the Chickpea Salad recipe are approximately 862 dietary Calories. Here's the breakdown of the ingredients and their calorie content:

Order ingredients

- 1 can chickpeas: 536 Calories
- 1 cucumber: 27 Calories
- 1 red bell pepper: 37 Calories
- 1/4 red onion: 11 Calories
- 1/4 cup fresh parsley: 5.5 Calories
- 2 tablespoons olive oil: 239 Calories
- 2 tablespoons lemon juice: 6.6 Calories

Now, let's order the ingredients on **Instacart**.

Used Instacart 

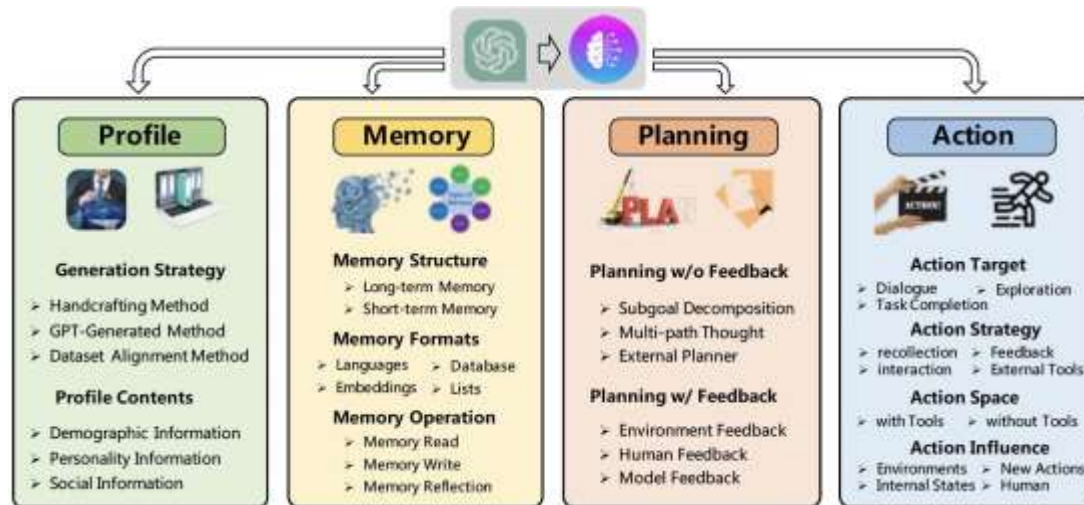
Here's your shopping list for the [Chickpea](#)

Project Background

April 2023: Autonomous Agents

"An autonomous agent is a system situated within and a part of an environment that senses that environment and acts on it, over time, in pursuit of its own agenda and so as to effect what it senses in the future."

Franklin and Graesser (1997)



```
Auto-GPT-stable — python • run.sh — 98x19
and language.']}
Continue (y/n): n
Welcome to Auto-GPT! run with '--help' for more information.
Create an AI-Assistant: Enter the name of your AI and its role below. Entering nothing will load defaults.
Name your AI: For example, 'Entrepreneur-GPT'
AI Name: AI agent writer
+AI agent writer here! I am at your service.
Describe your AI's role: For example, 'an AI designed to autonomously develop and run businesses with the sole goal of increasing your net worth.'
+AI agent writer is: Meant to search interesting news and important events about AI, technology and machine learning and create a summarized text. A final file with engaging tweets should be created as well.
Enter up to 5 goals for your AI: For example: Increase net worth, Grow Twitter Account, Develop and manage multiple businesses autonomously'
Enter nothing to load defaults, enter nothing when finished.
Goal 1: Look for news and articles about AI, technology and Data Science.
Goal 2: Summarize all in a well-structured and easy to read text.
Goal 3: 
```



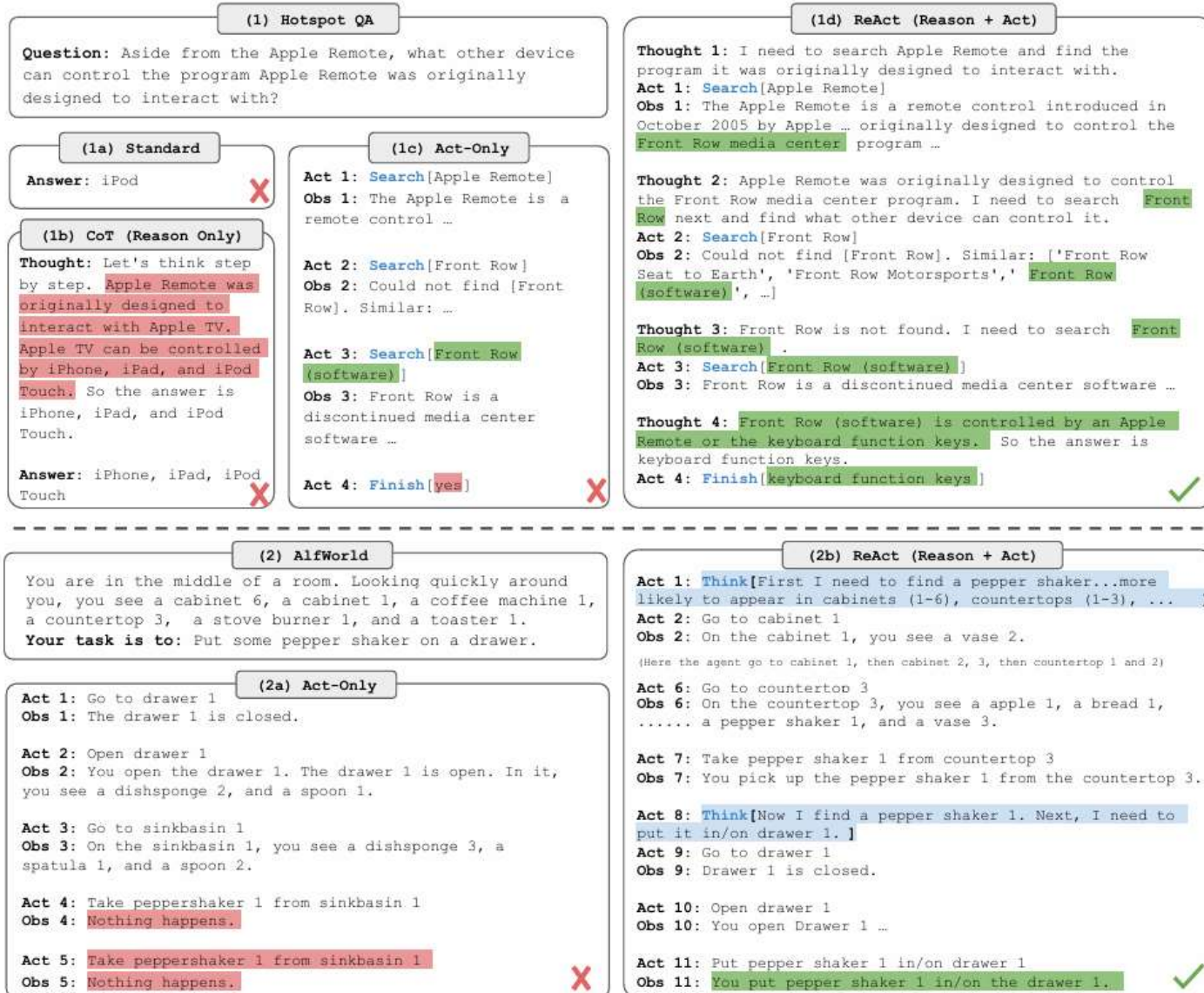
This is a pre-computed replay of a simulation that accompanies the paper entitled "Generative Agents: Interactive Simulacra of Human Behavior." It is for demonstration purposes only.

<https://arxiv.org/abs/2308.11432>

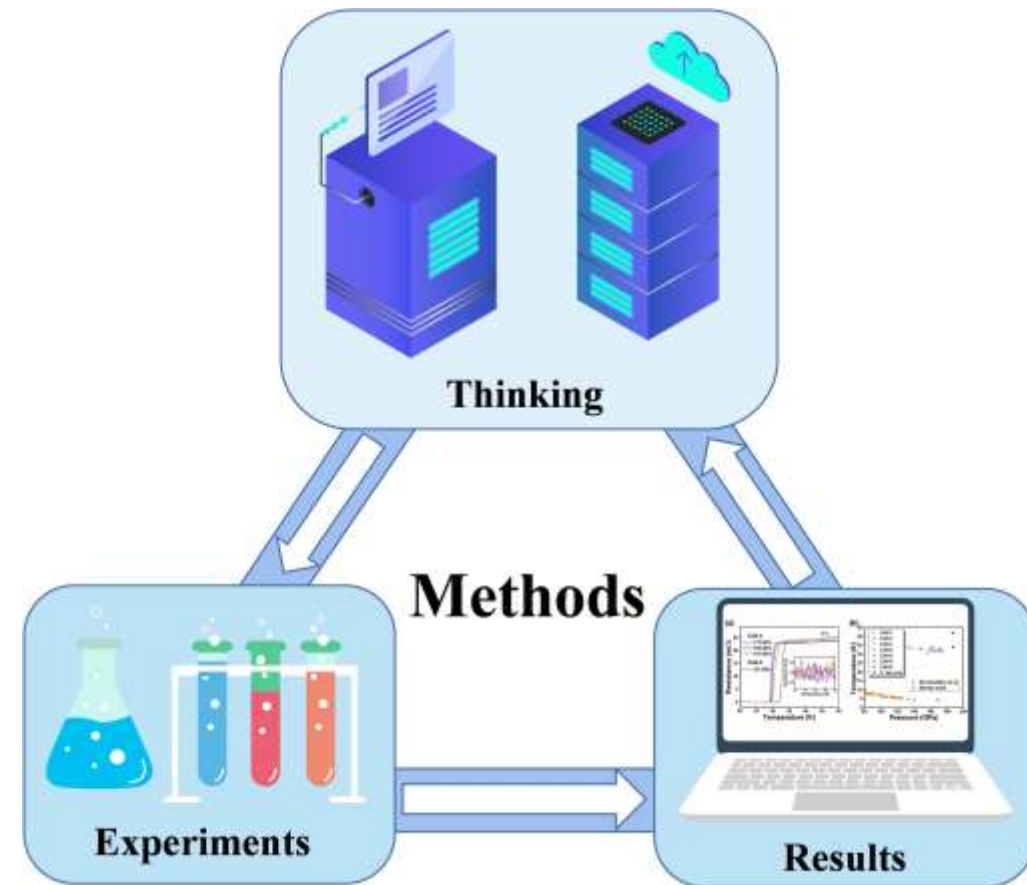
A Survey on Large Language Model based Autonomous Agents

Project Background

ReAct: The Synergy of Reasoning and Action in Language Models

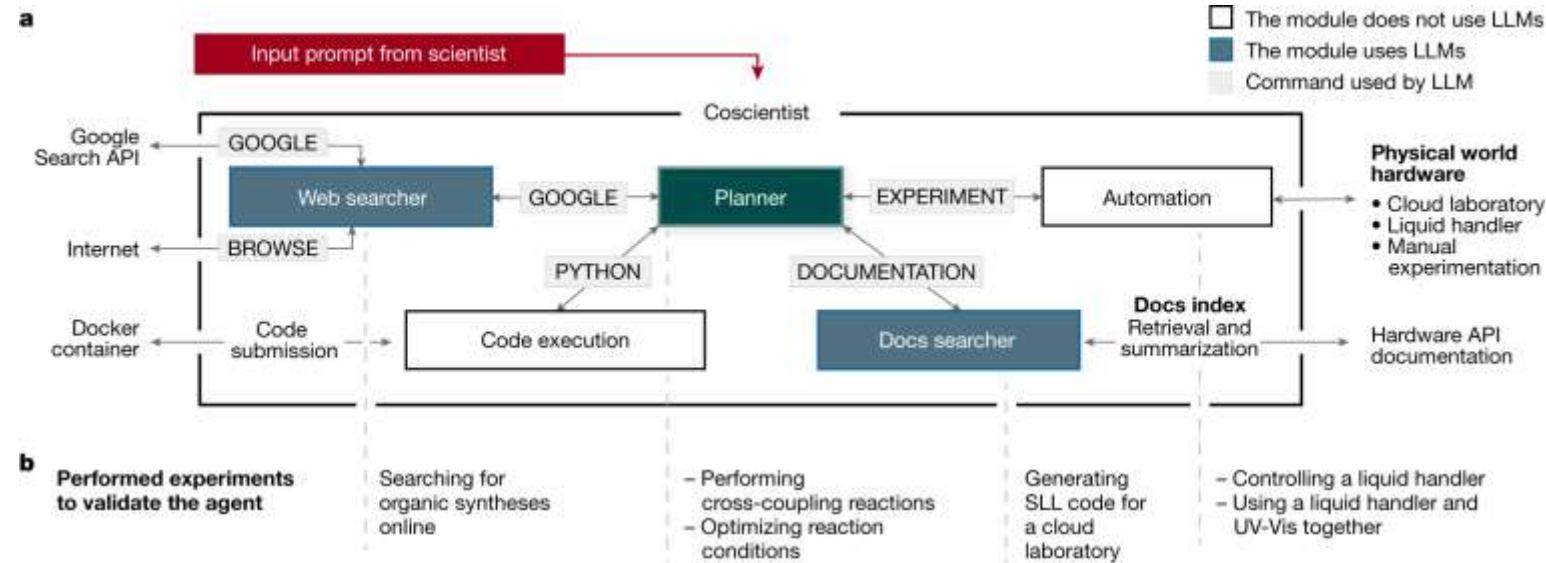


Experimental Disciplines



Project Background

AI Meets Chemistry: Turning the "Impossible" into the "Possible"

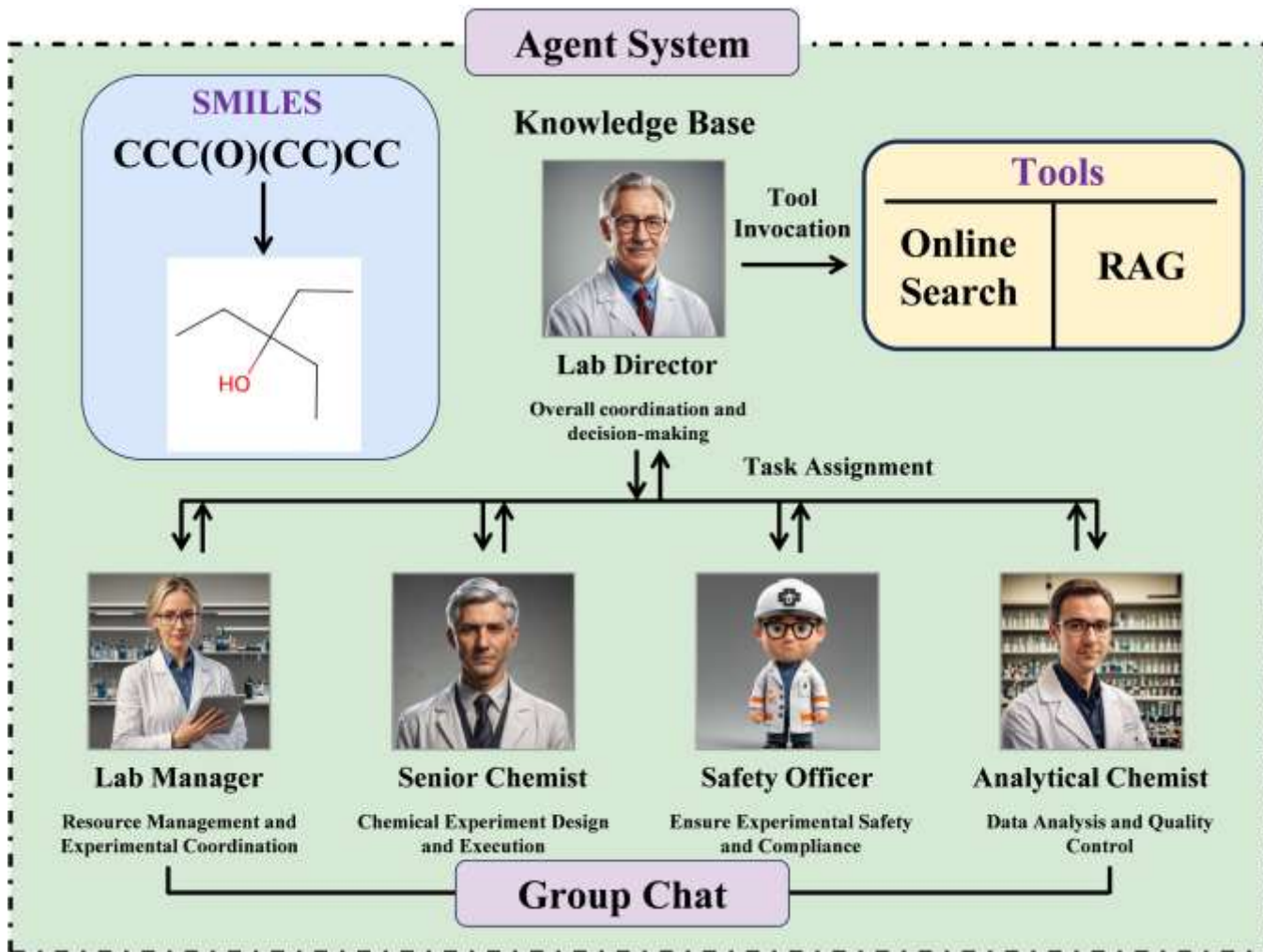


1. Online synthesis route planning
2. Search and organize literature
3. Cloud laboratory instructions
4. Precise control of reagent addition
5. Complete complex synthesis
6. Result analysis and optimization

Basic Functions of GVIM

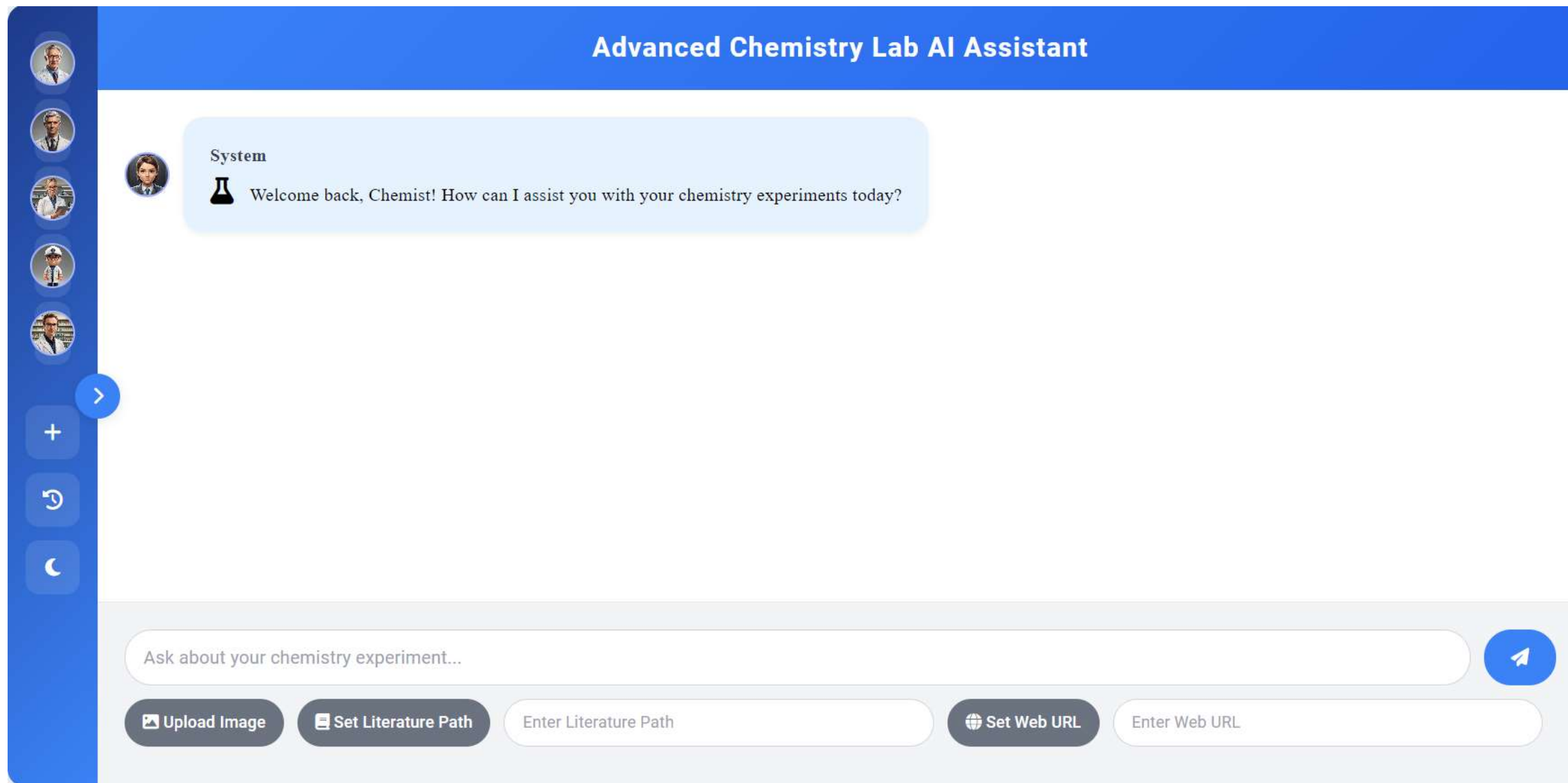
Basic Functions

Multi-Agent Collaboration



- **Multi-agent collaboration** simulates real human chemistry teams, providing multiple perspectives on chemical problems and proposing comprehensive and feasible solutions.
- **Online search and local knowledge base (RAG)** functionality (supports specified URL search and local document uploads).
- Use of **multimodal large models** to recognize handwritten chemical formulas.
- Visualization of SMILES characters in **3D and 2D structures**.
- AI-controlled browser functionality.
- Flexible use of **fine-tuned** or **non-fine-tuned** models.

Basic Functions



Basic Operation Interface

Basic Functions

CHEMISTRY TEAM

**Lab Director**
Level 3

**Senior Chemist**
Level 1

**Lab Manager**
Level 3

**Safety Officer**
Level 1

**Analytical Chemist**
Level 2

CHAT CONTROLS

+ New Experiment

 Past Experiments

Advanced Chemistry Lab AI Assistant

**System**
 Welcome back, Chemist! How can I assist you with your chemistry experiments today?

**You**
Please tell me some information about the journal Digital Discovery.

chat manager
Please tell me some information about the journal Digital Discovery.
[WEB_SEARCH_SUMMARY:Based on the search for 'Please tell me some information about the journal Digital Discovery.', here are the key findings: 1. [Digital Discovery Journal - The Royal Society of Chemistry](#) Article processing charges will apply to all articles submitted to Digital Discovery from 1 July 2024 onwards provided that, following peer review, they are accepted for

Ask about your chemistry experiment...

 Upload Image

 Set Literature Path

 Set Web URL



Online Search Function

Basic Functions

The screenshot displays the 'Advanced Chemistry Lab AI Assistant' interface. On the left is a sidebar for the 'CHEMISTRY TEAM' with five members: Lab Director (Level 2), Senior Chemist (Level 1), Lab Manager (Level 3), Safety Officer (Level 3), and Analytical Chemist (Level 2). Below this is the 'CHAT CONTROLS' section with buttons for '+ New Experiment' and 'Past Experiments'. The main area is titled 'Advanced Chemistry Lab AI Assistant' and features a 'Search Results' section. A modal window is open, displaying a search result for 'Solubility Rules for Ionic Compounds - MilliporeSigma'. The modal contains a paragraph about solubility and a link to the Sigma-Aldrich website. At the bottom of the main area is a search bar with the placeholder text 'Ask about your chemistry experiment...' and three buttons: 'Upload Image', 'Set Literature Path', and 'Set Web URL'.

CHEMISTRY TEAM

- Lab Director**
Level 2
- Senior Chemist**
Level 1
- Lab Manager**
Level 3
- Safety Officer**
Level 3
- Analytical Chemist**
Level 2

CHAT CONTROLS

- + New Experiment
- Past Experiments

Advanced Chemistry Lab AI Assistant

Search Results

Solubility Rules for Ionic Compounds - MilliporeSigma

Pressure and temperature affect solubility. This page discusses the solubility of compounds in water at room temperature and standard pressure. A compound that is soluble in water forms an aqueous solution. ... It can be helpful to write out the empirical formula so you can identify the ions that make up the compound.

<https://www.sigmaaldrich.com/GB/en/technical-documents/technical-article/materials-science-and-engineering/solid-state-synthesis/solubility-rules-solubility-of-common-ionic-compounds>

Ask about your chemistry experiment...

Upload Image Set Literature Path Set Web URL

Online Search Result Preview

Basic Functions

CHEMISTRY TEAM

**Lab Director**
Level 1
Leadership, Project Management

**Senior Chemist**
Level 1
Organic Chemistry, Inorganic Chemistry

**Lab Manager**
Level 1
Resource Management, Safety Protocols

**Safety Officer**
Level 1

CHAT CONTROLS

+ New Experiment

 Past Experiments

Advanced Chemistry Lab AI Assistant

**System**
 Welcome back, Chemist! How can I assist you with your chemistry experiments today?

Ask about your chemistry experiment...



 Upload Image

 Set Literature Path

 Set Web URL

Specified URL Search

Basic Functions

**Senior Chemist**
Level 2

**Lab Manager**
Level 1

**Safety Officer**
Level 1

**Analytical Chemist**
Level 3

CHAT CONTROLS

 New Experiment

 Past Experiments

 Toggle Theme

Advanced Chemistry Lab AI Assistant

**Lab_Director**

**Lab_Manager**

The tool has yielded a result. Based on this result, I will provide a direct response. The water solubility of CCC(O)CC is 0.14125375446227545 mol/L.

**Safety_Officer**

From a chemical safety perspective, it's essential to consider the handling and storage of CCC(O)CC, given its moderate water solubility. I recommend ensuring proper ventilation when handling this substance, wearing appropriate personal protective equipment (PPE), and storing it in a well-ventilated area to minimize exposure risks. Additionally, it's crucial to follow established protocols for waste disposal to prevent environmental contamination.



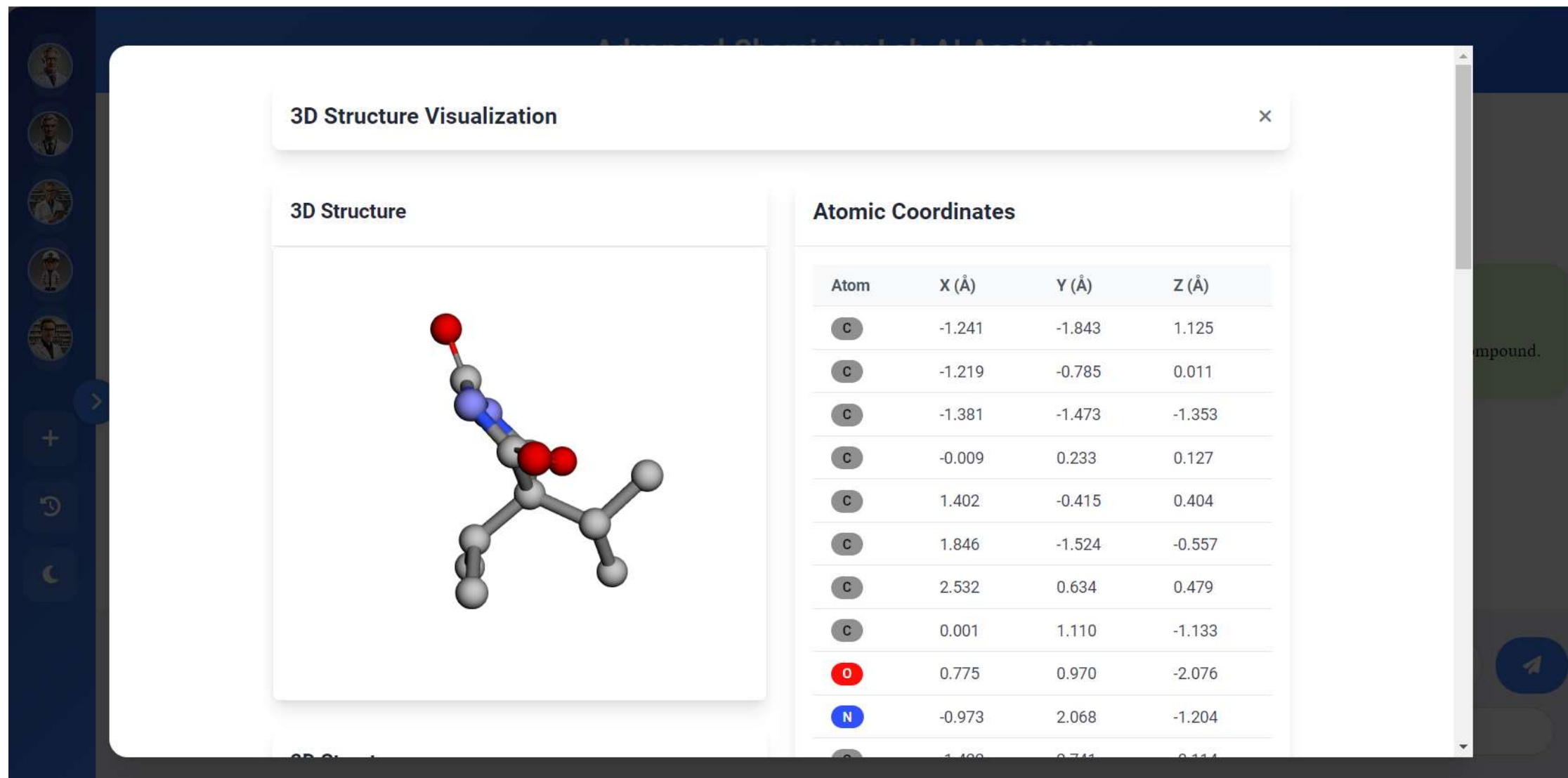
 Upload Image

 Set Literature Path

 Set Web URL

RAG Based on Local Knowledge Base

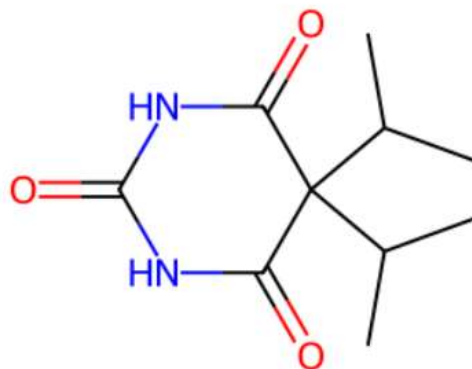
Basic Functions



SMILES Character 3D Structure Visualization

Basic Functions

2D Structure



C	-1.422	2.741	-0.114
O	-1.862	3.882	-0.203
N	-1.272	2.131	1.090
C	-0.365	1.136	1.327
O	0.073	0.977	2.464
H	-2.168	-2.425	1.078
H	-1.201	-1.383	2.116
H	-0.412	-2.550	1.043
H	-2.149	-0.212	0.141
H	-1.521	-0.746	-2.159
H	-2.273	-2.111	-1.350
H	-0.535	-2.109	-1.614
H	1.367	-0.879	1.399
H	1.818	-1.208	-1.603
H	1.235	-2.422	-0.448
H	2.875	-1.832	-0.339

SMILES Character 2D Structure Visualization

Basic Functions



>

+

↶

☾

Molecular Properties

Formula
C10H16N2O3

Molecular Weight
212.12 g/mol

LogP
0.65

TPSA
75.27 Å²

Rings
1

Rotatable Bonds
2

Structure Analysis

-  LogP indicates moderate lipophilicity
-  TPSA suggests limited membrane permeability
-  2 rotatable bonds indicate moderate flexibility
-  Contains 1 ring in the structure
-  Molecular weight: Moderate

Compound.



Introduction to Molecular Properties

Basic Functions

CHEMISTRY TEAM

**Lab Director**
Level 2

**Senior Chemist**
Level 2

**Lab Manager**
Level 1

**Safety Officer**
Level 1

**Analytical Chemist**
Level 3

CHAT CONTROLS

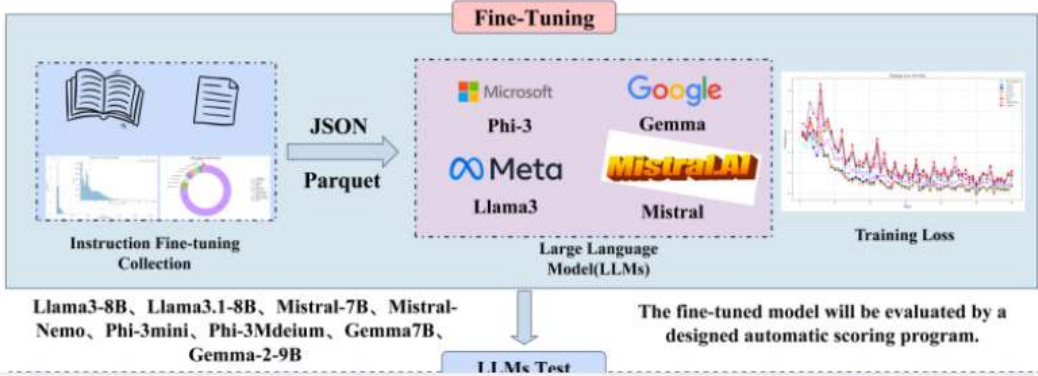
+ New Experiment

🕒 Past Experiments

Advanced Chemistry Lab AI Assistant

**System**
 Welcome back, Chemist! How can I assist you with your chemistry experiments today?

You
Please describe this picture.



The diagram illustrates the fine-tuning process for Large Language Models (LLMs). It starts with an 'Instruction Fine-tuning Collection' containing various documents and charts. This collection is processed into 'JSON' and 'Parquet' formats. These are then used for 'Fine-Tuning' on a 'Large Language Model (LLMs)'. The models listed include Microsoft Phi-3, Google Gemma, Meta Llama3, and Mistral AI Mistral. A 'Training Loss' graph is shown on the right. Below the diagram, a list of models is provided: Llama3-8B, Llama3.1-8B, Mistral-7B, Mistral-Nemo, Phi-3mini, Phi-3Mdeium, Gemma7B, and Gemma-2-9B. A note states: 'The fine-tuned model will be evaluated by a designed automatic scoring program.' Below the diagram is a button labeled 'LLMs Test'.

 Upload Image  Set Literature Path  Set Web URL

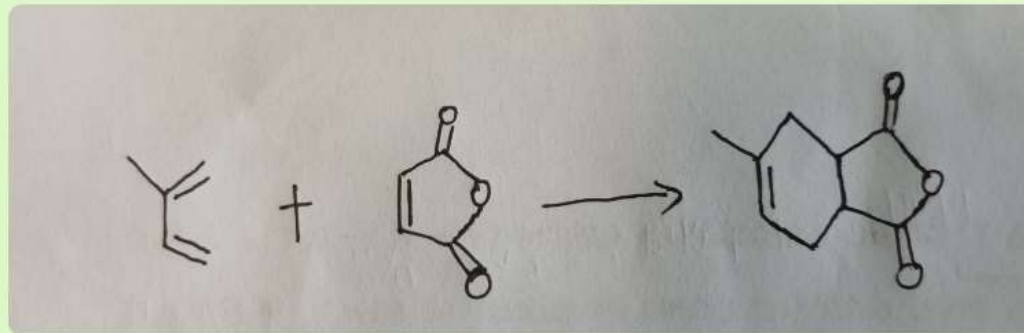


Image Recognition Description

Basic Functions

Advanced Chemistry Lab AI Assistant

Could you confirm if this handwritten chemical equation represents a Diels-Alder reaction?



chat_manager

Image Analysis: Yes, the handwritten chemical equation represents a Diels-Alder reaction. The Diels-Alder reaction is a chemical reaction between a conjugated diene and a substituted alkene, which results in the formation of a cyclohexene. In the image, the chemical equation shows a diene and an alkene reacting, leading to the formation of a cyclohexene. This confirms that the handwritten equation represents a Diels-Alder reaction. Could you confirm if this handwritten chemical

Ask about your chemistry experiment...

Upload Image

Set Literature Path

Set Web URL

Handwritten Chemical Structure Recognition

Basic Functions

CHEMISTRY TEAM

Lab Director

Level 1

Leadership, Project Management

Senior Chemist

Level 1

Organic Chemistry, Inorganic Chemistry

Lab Manager

Level 1

Resource Management, Safety Protocols

Safety Officer

Level 1

CHAT CONTROLS

+ New Experiment

Past Experiments

Advanced Chemistry Lab AI Assistant

Lab_Manager

It seems like the tool didn't quite understand what you were looking for. It's asking for more context or clarification on your question about chemical reactions with double bonds. Could you please rephrase or provide more information about what you're trying to find? Are you looking for a specific type of reaction, or perhaps a reaction mechanism that involves a double bond?

Safety_Officer

Well Protected

Good safety measures in place

Senior_Chemist

Pure Precision

Perfect management and coordination

Ask about your chemistry experiment...

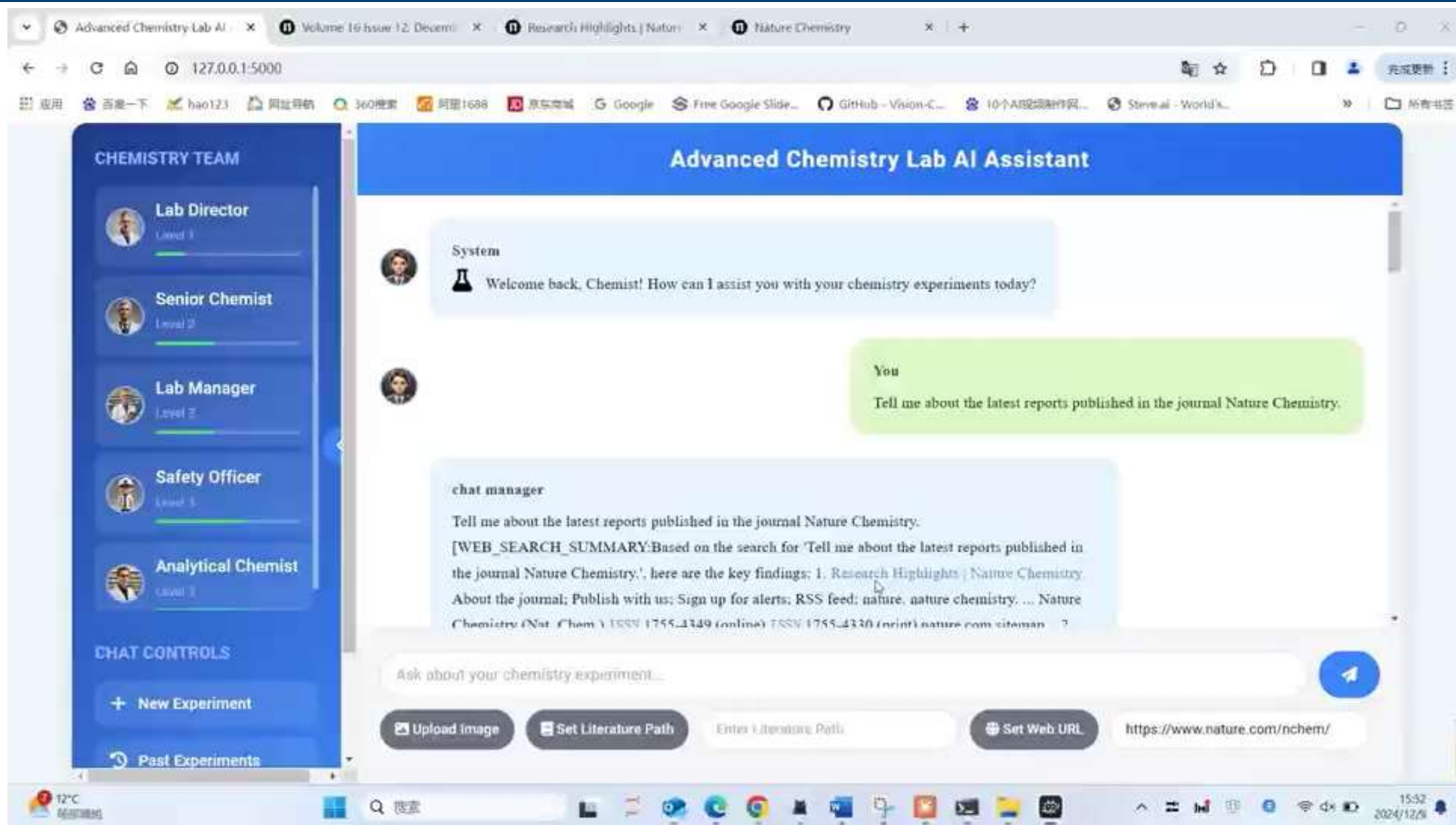
Upload Image

Set Literature Path

Set Web URL

User Feedback

Demo of GVIM



Preliminary Results



Showcasing research from Kangyong Ma's laboratory, College of Physics and Electronic Information Engineering, Zhejiang Normal University, Zhejiang, China.

AI agents in chemical research: GVIM – an intelligent research assistant system

GVIM is a multi-agent-based chemical research system that operates through the collaborative work of professional role agents such as laboratory directors and senior chemists. This research also fine-tunes large language models using over 1.7 million collected chemistry domain instruction data points, enabling the system to provide professional chemistry model selection. GVIM can flexibly utilize different large language models, and combined with retrieval-augmented generation and chemical structure visualization capabilities, it provides new development ideas for chemical research.

As featured in:



See Kangyong Ma, *Digital Discovery*, 2025, 4, 355.

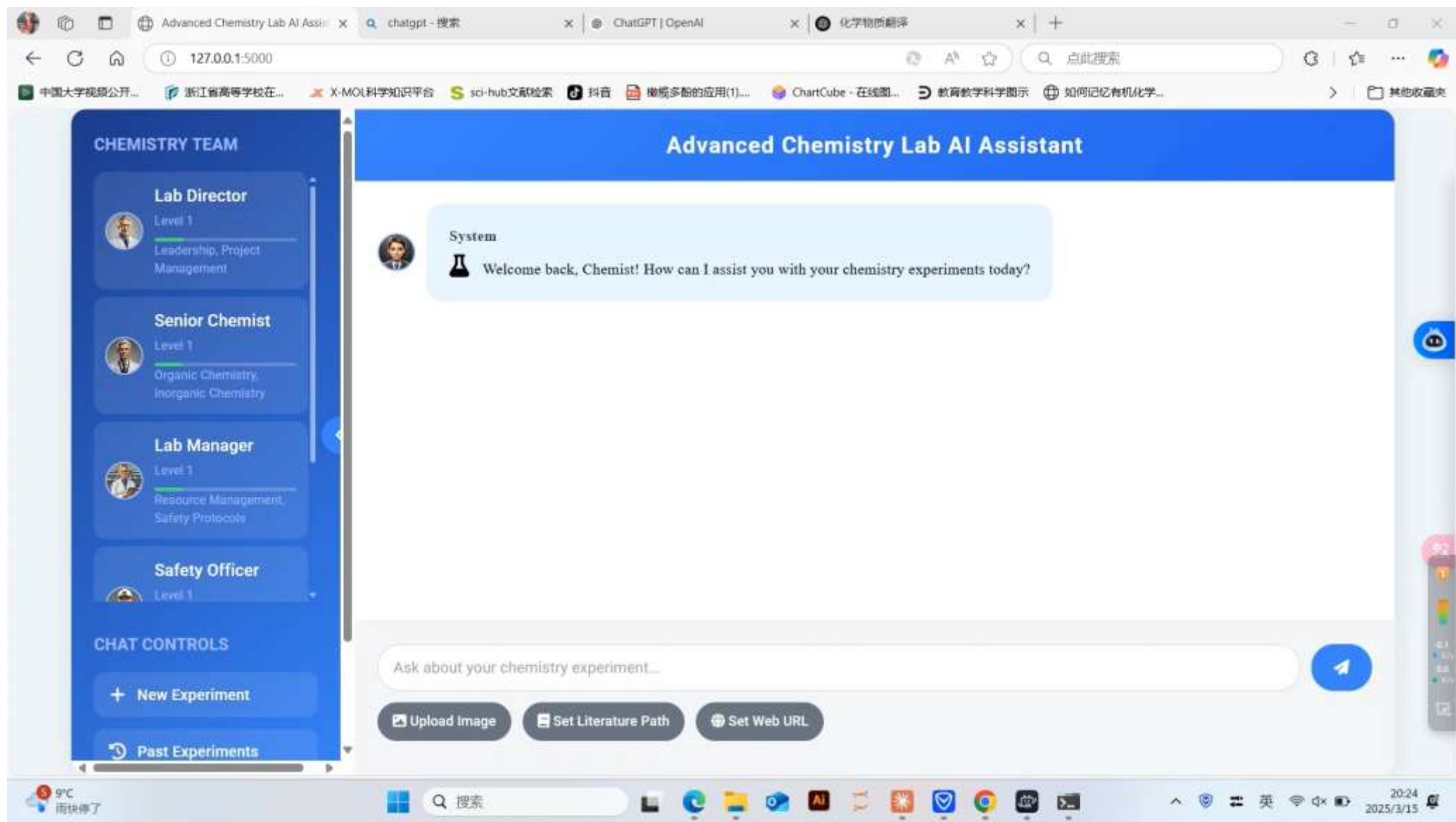


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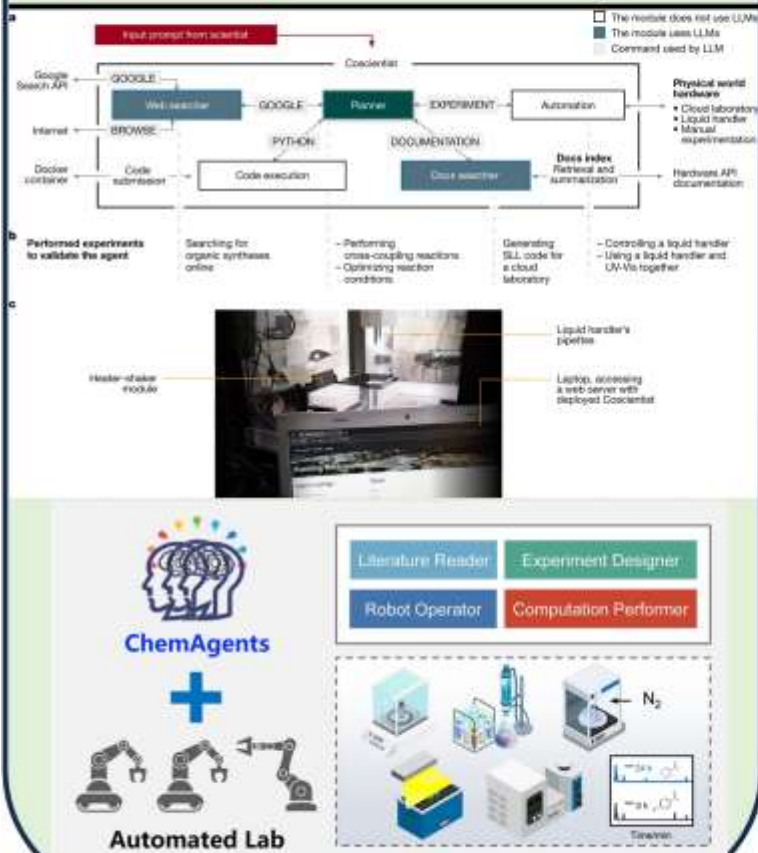
Future Prospects



AI-Controlled Browser Functionality

Future Prospects

Chemical Agents



MCP

BIOVIA | Materials Studio
Version 8
Modeling and Simulation Solutions for Chemicals and Materials Research

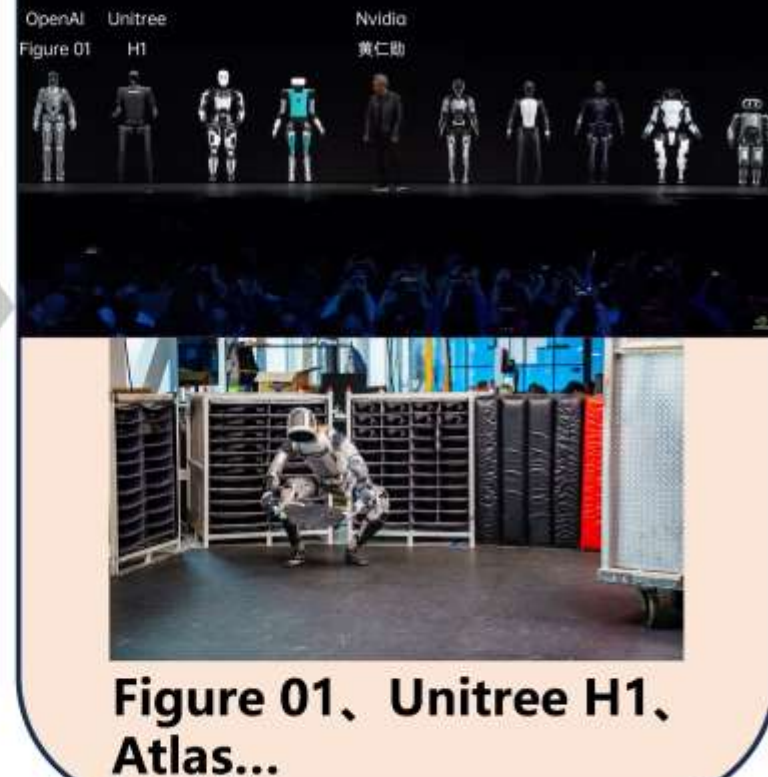
COM Server Setup
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PubChem
Molecule, atom, reaction, and multimolecule properties from molecular graphs

chemprop
MPNN → Property
FFN → Property

Software, Databases, Prediction Models ...

Embodied Intelligence



Chemical Research

Comprehensive Automation

High Intelligence